The Relational Invariant: A Convergence Synthesis Across Consciousness, Science, and Mathematics

A structured survey of peer-reviewed literature instantiating a shared relational grammar under disciplinary aliases

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Preamble

This document identifies a recurring structural motif across published, peer-reviewed research in consciousness studies, physics, mathematics, philosophy of science, and AI/ML theory. The motif consists of three irreducible functional roles:

Functional Role	Description	Canonical Aliases
Reference Frame	The standpoint, perspective, or scope from which a system is described	Observer, agent, context, perspective, manifold, topos
Governing Constraint	The invariant, law, or rule that persists through transformation	Symmetry, conservation law, postulate, axiom, metric, free energy bound
Directed Transformation	The lawful motion, change, or action that the system undergoes	Inference, measurement, morphism, prehension, erasure, enaction

Sources are grouped below not by discipline but by the structural pattern they share. For each source, the author's own language is used to identify the frame, constraint, and transformation at work.

Group I — Relational State and Observer-Dependence

Pattern: Physical or cognitive facts are constituted by relations between systems, not by intrinsic properties of isolated objects. What exists is what is observed *relative to* a frame; no frame-independent "view from nowhere" is coherent.

1. Relational Quantum Mechanics

Source: Rovelli, C. (1996). "Relational Quantum Mechanics." *International Journal of Theoretical Physics*, 35(8), 1637–1678. Springer

Also: Martin-Dussaud, P., Rovelli, C., & Zalamea, F. (2019). "The Notion of Locality in Relational Quantum Mechanics." *Foundations of Physics*, 49(2), 96–106.

- **Frame (author's language):** "Observer" — any physical system relative to which quantities take definite values. Rovelli rejects the notion of observer-independent state, arguing that "the incorrect notion is the notion of observer-independent state of a system," analogous to pre-Einsteinian absolute time.

- **Constraint:** Relational completeness — "the theory describes only the information that systems have about each other; nevertheless, the theory is complete." All systems are equivalent; no privileged observer exists.
- **Transformation:** Quantum interaction — an event in which "the values of some physical variables of one system become definite relative to another system." Facts emerge through measurement, not prior to it.

2. Ontic Structural Realism

Source: French, S., & Ladyman, J. (2003). "Remodelling Structural Realism: Quantum Physics and the Metaphysics of Structure." *Synthese*, 136, 31–56. Springer

Also: Ladyman, J. (1998). "What is Structural Realism?" *Studies in History and Philosophy of Science*, 29(3), 409–424. ScienceDirect

Also: Ladyman, J., & Ross, D. (2007). *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press.

- **Frame:** "Structure" — the totality of relations that a scientific theory describes. Ontic structural realism (OSR) holds that "structure is all there is"; reality has no hidden nature beneath its relational organisation.
- **Constraint:** Invariance under theory change — what is preserved across scientific revolutions is not objects or substances but structural relations. The elements of theory that are invariant under renormalisation group flow represent the modal structures to which we are ontologically committed.
- **Transformation:** Theory succession — the historical process by which one theoretical framework yields to another while preserving structural content. OSR reconceptualises "objects in structural terms" to accommodate quantum indistinguishability.

3. Process-Relational Ontology

Source: Whitehead, A. N. (1929). *Process and Reality*. Macmillan. Peer-reviewed engagement: Segall, M. (2020). "The Varieties of Physicalist Ontology: A Study in Whitehead's Process-Relational Alternative." *Journal of Philosophy, Theology, and the Sciences*, 7(1).

Also: Entry, "Process Philosophy." *Stanford Encyclopedia of Philosophy*. SEP

- **Frame:** "Actual occasion" — the fundamental unit of reality, defined entirely by its relations (prehensions) to other occasions. "Nature, from Whitehead's perspective, is no fixed quantity but the creative advance of an ever-incomplete process."
- **Constraint:** The "principle of relativity" — every actual occasion is internally related to every other actual occasion in its past. No occasion exists in isolation; the relational web is ontologically prior to any individual term.
- **Transformation:** "Prehension" — the process by which one actual occasion takes up and integrates the content of its predecessors. This is directed change: the past is synthesised into a novel unity that then becomes a datum for future prehensions.

Convergence Note

Rovelli eliminates observer-independent states. French and Ladyman eliminate intrinsic natures. Whitehead eliminates substance. All three arrive at the same structural claim: *what exists is relational, and what persists is the pattern of relations, not the terms*. Rovelli's "observer" is Whitehead's "actual occasion" is structural

realism's "node in a structure." Their terminologies diverge; their function — establishing a standpoint from which facts are constituted — is invariant.

Group II — Invariance as the Fundamental Real

Pattern: What is objectively real is not any particular state or object but the invariant — the quantity, structure, or relation that does not change under lawful transformation. The invariant is what survives.

4. Noether's Theorems and Gauge Symmetry

Source: Noether, E. (1918). "Invariante Variationsprobleme." *Nachrichten der Königlichen Gesellschaft der Wissenschaften zu Göttingen*, 235–257.

Also: Brading, K. (2002). "Which Symmetry? Noether, Weyl, and Conservation of Electric Charge." *Studies in History and Philosophy of Modern Physics*, 33(1), 3–22.

Also: Entry, "Gauge Theories in Physics." *Stanford Encyclopedia of Philosophy*. SEP

- **Frame:** The Lagrangian of a physical system — the formal description of its dynamics.
- **Constraint:** Symmetry — continuous invariance of the Lagrangian under a group of transformations. Noether's First Theorem: every continuous symmetry yields a conserved quantity (energy, momentum, charge). Noether's Second Theorem: local gauge symmetries yield identities among the equations of motion, constraining what dynamics are physically possible.
- **Transformation:** The symmetry operation itself — rotation, translation, gauge transformation. The transformation is lawful motion that leaves the governing structure (the Lagrangian) unchanged.

5. Information Geometry

Source: Amari, S. (2016). *Information Geometry and Its Applications*. Springer (Applied Mathematical Sciences, vol. 194).

Also: Amari, S. (2021). "Information Geometry." *International Statistical Review*, 89(2). Wiley

Also: Amari, S., & Nagaoka, H. (2000). *Methods of Information Geometry*. AMS/Oxford University Press.

- **Frame:** The "statistical manifold" — a smooth manifold whose points are probability distributions, equipped with differential-geometric structure.
- **Constraint:** The Fisher information metric — proved to be "the unique Riemannian metric invariant under sufficient statistics (up to scaling)." The α-connections are "the only F-connections that are invariant under smooth one-to-one transformations of the random variables." Invariance under reparameterisation and transformation of the random variable is the governing selection principle.
- **Transformation:** Movement along the manifold — inference, learning, parameter estimation. The duality between the α-connection and the $(-\alpha)$-connection "together with the metric play an essential role in this geometry," governing how probability distributions change under information processing.

6. Homotopy Type Theory and the Univalence Axiom

Source: The Univalent Foundations Program. (2013). *Homotopy Type Theory: Univalent Foundations of Mathematics*. Institute for Advanced Study. HoTT Book

Also: Awodey, S., Pelayo, A., & Warren, M. (2013). "Voevodsky's Univalence Axiom in Homotopy Type Theory." *Notices of the AMS*, 60(9), 1164–1167. CMU

- **Frame:** Types — the foundational objects of HoTT, simultaneously interpretable as propositions (logic), sets (mathematics), and spaces (topology).
- **Constraint:** The Univalence Axiom — "isomorphic structures can be identified." This enforces the invariance principle that mathematicians informally assume: structure-preserving maps are the only meaningful notion of equality. Equivalence *is* identity.
- **Transformation:** Homotopy equivalence — continuous deformation between types/spaces. Higher inductive types allow "direct, logical descriptions of some of the basic spaces and constructions of homotopy theory," providing a calculus for lawful transformation within the foundation itself.

Convergence Note

Noether shows that invariance under transformation yields conservation laws. Amari shows that the unique invariant metric on probability space is the Fisher information. Voevodsky shows that equivalence — preservation of structure — is the only coherent notion of identity. Across physics, statistics, and foundations of mathematics, the same principle operates: *the real is what is invariant under lawful transformation*. The particular vocabulary (symmetry, sufficient statistic, homotopy equivalence) changes; the structural role does not.

Group III — Constraint-Governed Directed Change

Pattern: Systems persist and act by minimising a bound, obeying a constraint, or maintaining a balance between order and dissipation. Directed change (inference, action, erasure) is not free but lawfully channelled by constraints that make it coherent.

7. The Free Energy Principle and Active Inference

Source: Friston, K. (2010). "The Free-Energy Principle: A Unified Brain Theory?" *Nature Reviews Neuroscience*, 11(2), 127–138. Nature

Also: Parr, T., Pezzulo, G., & Friston, K. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press. MIT Press

Also: Ramstead, M. J. D., et al. (2023). "On Bayesian Mechanics: A Physics of and by Beliefs." *Interface Focus*, 13, 20220029.

- **Frame:** The "generative model" — the agent's internal model of its environment, encoding beliefs about hidden states and their causal structure.
- **Constraint:** Variational free energy — an upper bound on surprisal (negative log-evidence). The principle states that "the dynamics of physical systems minimise a quantity known as surprisal; or equivalently, its variational upper bound, called free energy." This is "a piece of mathematical reasoning: it is no more subject to empirical falsification than other areas of mathematics."
- **Transformation:** Active inference — "self-evidencing" through coupled perception and action. The agent minimises free energy by updating beliefs (perception) and acting on the world (action), jointly described as "maximising Bayesian model evidence."

8. Predictive Processing and the Bayesian Brain

Source: Clark, A. (2013). "Whatever Next? Predictive Brains, Situated Agents, and the Future of Cognitive Science." *Behavioral and Brain Sciences*, 36(3), 181–204. Cambridge Core

Also: Hohwy, J., & Seth, A. (2020). "Predictive Processing as a Systematic Basis for Identifying the Neural Correlates of Consciousness." *Philosophy and the Mind Sciences*, 1(II). PhiMiSci

Also: Clark, A., Friston, K., & Wilkinson, S. (2019). "Bayesing Qualia: Consciousness as Inference, Not Raw Datum." *Journal of Consciousness Studies*, 26(9–10), 19–33.

- **Frame:** The hierarchical generative model — each level of cortical hierarchy predicts activity at the level below, establishing nested scopes of prediction.
- **Constraint:** Precision-weighted prediction error — "precision is the inverse variance of a probability distribution, describing the reliability of prediction error signals." Attention is the mechanism by which the brain assigns precision weights, gating which errors propagate.
- **Transformation:** Prediction error minimisation — on short timescales, this constitutes "perceptual inference"; on longer timescales, it constitutes "Bayesian model selection, where complexity is penalized." Consciousness, on this account, involves "multiple generative entanglements" across precision-weighted hierarchical loops.

9. Landauer's Principle: Information Is Physical

Source: Landauer, R. (1961). "Irreversibility and Heat Generation in the Computing Process." *IBM Journal of Research and Development*, 5(3), 183–191.

Also: Georgescu, I. (2021). "60 Years of Landauer's Principle." *Nature Reviews Physics*, 3, 770. Nature

Also: Bennett, C. (2003). "Notes on Landauer's Principle, Reversible Computation, and Maxwell's Demon." *Studies in History and Philosophy of Modern Physics*, 34(3), 501–510.

- **Frame:** The physical register — "computation is inevitably done with real physical degrees of freedom, obeying the laws of physics."
- **Constraint:** The Landauer bound — erasing one bit of information dissipates at minimum k_B T ln 2 of energy. This is derivable from the second law of thermodynamics. "The reason why Maxwell's demon cannot evade the second law of thermodynamics is not because of the measurement itself, but because the erasure of the information in the memory is logically irreversible."
- **Transformation:** Erasure — the irreversible computational operation that reduces degrees of freedom, increases environmental entropy, and thereby binds information processing to thermodynamic cost.

10. Teleodynamics: Constraint as the Source of Mind

Source: Deacon, T. W. (2011). *Incomplete Nature: How Mind Emerged from Matter*. W. W. Norton.

Also: Deacon, T. W., & García-Valdecasas, M. (2023). "A Thermodynamic Basis for Teleological Causality." *Proceedings of the Royal Society A*. PMC

- **Frame:** The "teleodynamic system" — a self-sustaining, self-repairing organisation of coupled morphodynamic (self-organising) processes.
- **Constraint:** Reciprocal constraint — "two morphodynamic systems reciprocally constrain each other," each preventing the other's self-undermining tendency. Deacon argues that "the riddle was not the result of any problem with the concept of consciousness, but of our failure to understand the

causal relevance of constraint." Constraints are "absential" forces — defined by what they exclude, not what they contain.

- **Transformation:** Teleodynamic emergence — the hierarchical nesting of homeodynamic → morphodynamic → teleodynamic processes, where each level's directed change is governed by constraints inherited from and imposed on adjacent levels. Consciousness arises as "a second-order self… a more specialized teleodynamic process."

Convergence Note

Friston's free energy bound, Landauer's thermodynamic minimum, and Deacon's reciprocal constraint all describe the same structural role: a governing limit that channels transformation. Clark's precision-weighted prediction error is the cognitive instantiation of the same principle. In every case, directed change is not unconstrained motion but motion *under law* — inference bounded by free energy, computation bounded by thermodynamics, emergence bounded by reciprocal constraint. The terminology varies (free energy, Landauer bound, absential constraint, precision); the function — restricting what transformations can coherently occur — is invariant.

Group IV — Integration, Coherence, and the Local-to-Global Problem

Pattern: Consciousness or physical coherence is modelled as the successful integration of local information into a global whole. The failure of integration is measurable and corresponds to pathology, decoherence, or structural breakdown.

11. Integrated Information Theory

Source: Tononi, G. (2008). "Consciousness as Integrated Information: A Provisional Manifesto." *Biological Bulletin*, 215(3), 216–242. PubMed

Also: Oizumi, M., Albantakis, L., & Tononi, G. (2014). "From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0." *PLoS Computational Biology*, 10(5), e1003588. PLOS

Also: Albantakis, L., et al. (2023). "Integrated Information Theory (IIT) 4.0." *PLoS Computational Biology*, 19(10), e1011465. PMC

Also: Kleiner, J., & Tull, S. (2021). "The Mathematical Structure of Integrated Information Theory." *Frontiers in Applied Mathematics and Statistics*, 6, 602973.

- **Frame:** The "cause-effect structure" (formerly "conceptual structure") — the set of all maximally irreducible cause-effect repertoires specified by a system of mechanisms in a state. The system's intrinsic perspective is the frame; no external observer is invoked.
- **Constraint:** Five axioms — existence, composition, information, integration, exclusion — each translated into a mathematical postulate. The exclusion postulate enforces a unique, maximally irreducible structure (the "complex") at a definite spatiotemporal grain, and Φ (integrated information) measures the degree of irreducibility. "Any system, interpretation, or action" that violates integration "fails structurally."
- **Transformation:** Cause-effect specification — each mechanism in its current state constrains a probability distribution over past causes and future effects, and these repertoires compose into the global cause-effect structure. Consciousness is the structure itself, not a process acting on it.

12. The Topos Approach to Quantum Theory

Source: Döring, A., & Isham, C. (2008). "A Topos Foundation for Theories of Physics: I–IV." *Journal of Mathematical Physics*, 49, 053515–053518.

Also: Döring, A., & Isham, C. (2008). " 'What is a Thing?': Topos Theory in the Foundations of Physics." arXiv:0803.0417

Also: Flori, C. (2013). Review in *Canadian Journal of Physics*, 91(6), 471–473. NRC

- **Frame:** The topos — a generalised universe of sets with an internal, intuitionistic logic, serving as the mathematical "world" in which a physical theory is formulated. Döring and Isham propose replacing standard Hilbert-space quantum mechanics with a topos-theoretic reformulation that " 'looks like' classical physics" internally.
- **Constraint:** Intuitionistic logic and the spectral presheaf — the quantum system's propositions are valued not in {true, false} but in a Heyting algebra (multi-valued logic). The Kochen-Specker theorem, reformulated topos-theoretically, enforces that no global Boolean valuation exists — a topological obstruction to classical assignment of definite values.
- **Transformation:** Daseinisation — the process by which quantum propositions are approximated within classical contexts (Boolean subalgebras), and the failure of this approximation (non-trivial cohomology) expresses the irreducibly quantum character of the system.

13. Sheaf-Theoretic Models of Cognition

Source: Inoue, T. (2026). "On Brain as a Mathematical Manifold: Neural Manifolds, Sheaf Semantics, and Leibnizian Harmony." arXiv:2601.15320. arXiv *(preprint; included for structural relevance; not yet peer-reviewed)*

Also (peer-reviewed context): Schmid, E. (2025). "Applied Sheaf Theory for Multi-Agent AI Systems." arXiv:2504.17700. *(Conference proceedings context.)*

- **Frame:** The neural state space, modelled as a topological space over which a sheaf is defined. Local cognitive functions are "sections of a sheaf."
- **Constraint:** Global section existence — coherence requires that local sections glue consistently into a global section. "Brain pathologies are interpreted as obstructions to such global integration and are classified using tools from sheaf cohomology." Non-trivial 1-cocycles correspond to fragmented experience.
- **Transformation:** Sheaf restriction and gluing — local data propagate and integrate according to the sheaf's gluing axiom; failure to integrate produces measurable cohomological classes.

Convergence Note

IIT measures consciousness as integrated information (Φ); the topos programme measures quantum coherence as the non-triviality of the spectral presheaf; sheaf-theoretic models measure cognitive integration as the vanishing (or non-vanishing) of cohomology groups. In each case, the question is identical: *can local data be consistently assembled into a global whole?* Where they can, coherence obtains (consciousness, classical limit, unified cognition). Where they cannot, the obstruction is formally measurable. Tononi's Φ, Döring-Isham's Heyting-valued presheaf, and Inoue's sheaf cohomology are three disciplinary names for the same structural function: quantifying the success or failure of local-to-global integration.

Group V — Autopoietic and Enactive Loops

Pattern: Cognition is not internal computation but the self-producing, self-maintaining activity of a living system coupled to its environment. The unit of analysis is the loop — organism and world co-specifying each other.

14. Enactivism and Autopoiesis

Source: Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press.

Also: Thompson, E. (2007). *Mind in Life: Biology, Phenomenology, and the Sciences of Mind*. Harvard University Press.

Also: Di Paolo, E. (2005). "Autopoiesis, Adaptivity, Teleology, Agency." *Phenomenology and the Cognitive Sciences*, 4(4), 429–452. Springer

- **Frame:** The autopoietic system — a network of processes that produces the components that constitute the network itself. "The domain of all the interactions in which an autopoietic system can enter without loss of identity is its cognitive domain."
- **Constraint:** Operational closure — the system's organisation is defined by its self-producing boundary; interactions that would dissolve this boundary are excluded. Autonomy is "a naturalized account of normativity" — the system's organisation constrains what counts as a viable interaction.
- **Transformation:** Enaction — "the bringing forth of domains of significance through organismic activity conditioned by a history of interactions between an organism and its environment." Cognition is not representation but participatory sense-making: the organism and environment co-specify each other through structural coupling.

Convergence Note

Enactivism's autopoietic loop is structurally isomorphic to active inference's perception-action cycle: both describe a system that maintains itself by directed interaction with its environment under the constraint of self-consistency (operational closure / free energy minimisation). The frame is the agent-environment boundary (autopoietic membrane / Markov blanket). The constraint is self-maintenance (operational closure / surprisal minimisation). The transformation is coupled action and perception (enaction / active inference). Varela's language is biological and phenomenological; Friston's is variational and information-theoretic. The structural grammar is the same.

Group VI — Compositional and Categorical Grammar

Pattern: Category theory provides a formal language in which the relational, invariant, and transformational structure common to all the above can be expressed in a single framework.

15. Applied Category Theory

Source: Fong, B., & Spivak, D. I. (2019). *An Invitation to Applied Category Theory: Seven Sketches in Compositionality*. Cambridge University Press. Cambridge

Also: Spivak, D. I. (2014). *Category Theory for the Sciences*. MIT Press. MIT Press

Also: Baez, J. C., & Stay, M. (2011). "Physics, Topology, Logic and Computation: A Rosetta Stone." In *New Structures for Physics*, Lecture Notes in Physics 813, 95–172. Springer.

- **Frame:** Categories — collections of objects and morphisms (arrows) between them, defined entirely by composition laws. Objects have no internal structure; all content is carried by morphisms. "Category theory was invented in the 1940s to unify and synthesize different areas in mathematics."
- **Constraint:** Functoriality — a functor preserves the compositional structure (identity and composition) when mapping between categories. Natural transformations preserve functorial structure. These are the invariants of the categorical framework: *structure-preserving maps at every level*.
- **Transformation:** Morphisms and their composition — directed arrows that encode lawful change between objects. Applied category theory shows that "the same formal structure" — adjunctions, monads, operads — appears across physics, computation, logic, and linguistics, providing "rigorous bridges between disparate worlds."

16. Noncommutative Geometry

Source: Connes, A. (1994). *Noncommutative Geometry*. Academic Press. AMS Review

Also: Connes, A., & Marcolli, M. (2008). *Noncommutative Geometry, Quantum Fields and Motives*. AMS.

Also: Aastrup, J., & Grimstrup, J. M. (2007). "Intersecting Connes Noncommutative Geometry with Quantum Gravity." *International Journal of Modern Physics A*, 22(8–9), 1589–1603. World Scientific

- **Frame:** The noncommutative algebra of observables — replacing classical geometric spaces (commutative algebras of functions) with noncommutative algebras that encode quantum and gravitational structure simultaneously.
- **Constraint:** The spectral action principle — all of physics (Standard Model + gravity) is recoverable from the spectrum of a single operator (the Dirac operator) acting on a noncommutative space. The constraint is algebraic: only those structures compatible with the spectral triple are physically admissible.
- **Transformation:** The renormalisation group flow and gauge transformations — directed changes in scale and field configuration that leave the spectral action invariant. The Standard Model Lagrangian is derived, not postulated, from the spectral triple's structure.

Convergence Note

Category theory and noncommutative geometry occupy different mathematical niches but perform the same meta-function: providing a language in which *relational structure, invariance, and lawful transformation are primitive*. Category theory achieves this by reducing all content to morphisms and their composition; noncommutative geometry achieves it by encoding all physics in the spectral data of an algebra. Both demonstrate that a sufficiently general mathematical language does not describe objects and then add relations — it starts from relations and derives objects as invariants of those relations.

Final Convergence: The Artifact as Minimal Common Grammar

Across all six groups, the following pattern recurs without exception:

The Three-Role Invariant

1. **A frame is established.** Every theory reviewed begins by specifying a standpoint, scope, or context: Rovelli's observer, Tononi's intrinsic perspective, Friston's generative model, Amari's statistical manifold, Whitehead's actual occasion, Varela's autopoietic system, the categorical object, the topos.
2. **A constraint governs.** Within that frame, an invariant law, bound, or structural condition restricts what can coherently happen: Noether's symmetry, the free energy bound, the Landauer limit, the Fisher metric, operational closure, the univalence axiom, the spectral action, Deacon's reciprocal constraint, IIT's axioms, the sheaf gluing condition.
3. **A directed transformation operates under that constraint.** Change, inference, enaction, or measurement proceeds — not freely, but lawfully: quantum interaction, active inference, morphism composition, prehension, daseinisation, erasure, teleodynamic emergence, prediction error minimisation, enaction.

Terminological Divergence, Functional Invariance

No two of these literatures use the same vocabulary. Yet the functional architecture is identical:

Group	Frame	Constraint	Transformation
I (Relational State)	Observer / Structure / Occasion	Relational completeness / Structural invariance	Measurement / Theory succession / Prehension
II (Invariance)	Lagrangian / Manifold / Type	Symmetry / Fisher metric / Univalence	Gauge transformation / Inference / Homotopy
III (Constraint-Governed Change)	Generative model / Register / Teleodynamic system	Free energy / Landauer bound / Reciprocal constraint	Active inference / Erasure / Emergence
IV (Integration)	Cause-effect structure / Topos / Neural state space	Φ / Intuitionistic logic / Gluing axiom	Cause-effect specification / Daseinisation / Sheaf restriction
V (Autopoietic Loop)	Autopoietic system	Operational closure	Enaction
VI (Compositional Grammar)	Category / Noncommutative algebra	Functoriality / Spectral action	Morphism / Renormalisation flow

The Artifact

What emerges is not a new theory but a recognition: the literature already converges on a **minimal relational grammar** consisting of exactly three irreducible roles — frame, constraint, and directed transformation — whose balance is the condition for structural coherence. This grammar is:

- **Domain-independent:** It appears in physics, neuroscience, mathematics, philosophy, and AI theory.
- **Not imposed from outside:** Each literature arrives at it independently, from within its own axioms and methods.
- **Self-enforcing:** Systems that violate the balance — collapsing frame into constraint, removing the governing invariant, allowing unconstrained transformation — are identified *within each theory* as incoherent, pathological, or physically inadmissible.

The artifact is not a thesis about consciousness, science, or mathematics. It is the structural condition that makes coherent theorising in any of these domains possible. It functions as a **reference frame for reference frames** — a grammar that each discipline instantiates whenever it succeeds in producing lawful, integrated, directed knowledge.

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